Prospects of immune checkpoint modulators in the treatment of glioblastoma.

PMID: 26260659 | Indexed in PubMed

Glioblastoma is the most common primary brain tumour in adults. Prognosis is poor: even with the current gold-standard first-line treatment—maximal safe resection and combination of radiotherapy with temozolomide chemotherapy—the median overall survival time is only approximately 15-17 months, because the tumour recurs in virtually all patients, and no commonly accepted standard treatment for recurrent disease exists. Several targeted agents have failed to improve patient outcomes in glioblastoma. Immunotherapy with immune checkpoint inhibitors such as ipilimumab, nivolumab, and pembrolizumab has provided relevant clinical improvements in other advanced tumours for which conventional therapies have had limited success, making immunotherapy an appealing strategy in glioblastoma. This Review summarizes current knowledge on immune checkpoint modulators and evaluates their potential role in glioblastoma on the basis of preclinical studies and emerging clinical data. Furthermore, we discuss challenges that need to be considered in the clinical development of drugs that target immune checkpoint pathways in glioblastoma, such as specific properties of the immune system in the CNS, issues with radiological response assessment, and potential interactions with established and emerging treatment strategies.

A patient-derived xenograft mouse platform from epithelioid glioblastoma provides possible druggable screening and translational study.

PMID: 39553226 | Indexed in PubMed

Despite advancements in targeted therapy, glioblastoma remains a challenging condition with limited treatment options. While surgical techniques and external radiation therapy have improved, the median survival for glioblastoma stands at around 12-18 months, with a 5-year survival rate of only 6.8%. Epithelioid glioblastoma (eGBM) represents a rare subtype within the glioma spectrum. Utilizing patient-derived xenograft (PDX) models in mice offers a promising avenue for drug screening and translational research, particularly for this specific glioblastoma subtype. Establishing a stable PDX model for eGBM revealed consistent genetic abnormalities, including BRAF V600E mutation and CDKN2A deletion, in both primary and PDX tumors. Leveraging a curated drug database, compounds potentially targeting these aberrations were identified. By using the novel PDX platform, the results presented in this study demonstrate that the treatments with Palbociclib or Dabrafenib/Trametinib significantly reduced tumor size. RNA sequencing analysis further validated the responsiveness of the tumors to these targeted therapies. In conclusion, PDX models offer a deeper understanding of eGBM at the genomic level and facilitate the identification of potential therapeutic targets. Further translational studies of this novel PDX model hold promise for advancing the diagnosis and treatment of this specific subtype of glioblastoma.

Phase 1b/2 study of orally administered pexidartinib in combination with radiation therapy and temozolomide in patients with newly diagnosed glioblastoma.

PMID: 39734810 | Indexed in PubMed

Glioblastoma (GBM) has a median survival of <2 years. Pexidartinib (PLX3397) is a small-molecule inhibitor of CSF1R, KIT, and oncogenic FTL3, which are implicated in GBM treatment resistance. Results from glioma models indicate that combining radiation therapy (RT) and pexidartinib reduces radiation resistance. We added pexidartinib to standard-of-care RT/temozolomide (TMZ) in patients with newly diagnosed GBM to assess the therapeutic benefit of altering the tumor microenvironment with pexidartinib. In this open-label, dose-escalation, multicenter, Phase 1b/2 trial, pexidartinib was administered in combination with RT/TMZ followed by adjuvant pexidartinib + TMZ. During Phase 1b, pexidartinib was given 5 or 7 days/week at multiple dosing levels. The primary Phase 1b endpoint was the recommended Phase 2 dose (RP2D). Phase 2 patients received the RP2D with the primary endpoint of median progression-free survival (mPFS). Secondary objectives were median overall survival (mOS), pharmacokinetics, and safety. The RP2D of pexidartinib was 800 mg/day for 5 days/week during RT/TMZ, followed by 800 mg/day for 7 days/week with adjuvant TMZ. mPFS was 6.7 months (90% CI: 4.5, 11.5) for the modified intention-to-treat population. The actual mOS was 13.1 months (90% CI: 11.5, 24.5), and the mOS corrected for comparison with matched historical controls was 18.8 months (95% CI: 12.6, 28.0). This trial established the RP2D of pexidartinib in combination with RT/TMZ and adjuvant TMZ. Pexidartinib was generally safe and well tolerated. Although the study regimen with pexidartinib was not efficacious, pharmacodynamic studies showed modulation of systemic markers that could lead to alteration of the tumor microenvironment.

Novel Fibroblast Growth Factor Receptor 3-Fatty Acid Synthase Gene Fusion in Recurrent Epithelioid Glioblastoma Linked to Aggressive Clinical Progression.

PMID: 39590169 | Indexed in PubMed

Glioblastoma (GBM) is the most common primary malignant brain tumor in adults, with a median overall survival (OS) of 15-18 months despite standard treatments. Approximately 8% of GBM cases exhibit genomic alterations in fibroblast growth factor receptors (FGFRs), particularly FGFR1 and FGFR3. Next-generation sequencing techniques have identified various FGFR3 fusions in GBM. This report presents a novel FGFR3 fusion with fatty acid synthase (FASN) in a 41-year-old male diagnosed with GBM. The patient presented with a persistent headache, and imaging revealed a right frontal lobe lesion. Surgical resection and subsequent histopathology confirmed GBM. Initial NGS analysis showed no mutations in the IDH1, IDH2 or H3F3 genes, but revealed a TERT promoter mutation and CDKN2A/2B and PTEN deletions. Postoperative treatment included radiotherapy and temozolomide. Despite initial management, recurrence occurred four months post-diagnosis, confirmed by MRI and histology. A second surgery identified a novel FGFR3-FASN fusion, alongside increased Ki67 expression. The recurrence was managed with regorafenib and bevacizumab, though complications like hand-foot syndrome and radiation necrosis arose. Despite initial improvement, the patient died 15 months after diagnosis. This case underscores the importance of understanding GBM's molecular landscape for effective treatment strategies. The novel FGFR3-FASN fusion suggests potential implications for GBM recurrence and lipid metabolism. Further studies are warranted to explore FGFR3-FASN's role in GBM and its therapeutic targeting.

Mitochondrial Dysfunction: Effects and Therapeutic Implications in Cerebral Gliomas.

PMID: 39597073 | Indexed in PubMed

Gliomas are the most common primary brain tumors, representing approximately 28% of all central nervous system tumors. These tumors are characterized by rapid progression and show a median survival of approximately 18 months. The therapeutic options consist of surgical resection followed by radiotherapy and chemotherapy. Despite the multidisciplinary approach and the biomolecular role of targeted therapies, the median progression-free survival is approximately 6-8 months. The incomplete tumor compliance with treatment is due to several factors such as the presence of the blood-brain barrier, the numerous pathways involved in tumor transformation, and the presence of intra-tumoral mutations. Among these, the interaction between the mutations of genes involved in tumor bio-energetic metabolism and the functional response of the tumor has become the protagonist of numerous studies. In this scenario, the main role is played by mitochondria, cellular organelles delimited by a double membrane and containing their own DNA (mtDNA), which participates in numerous cellular processes such as the regulation of cellular metabolism, cellular proliferation, and apoptosis and is also the main source of cellular energy production. Therefore, it is understood that the mitochondrion, specifically its functional alteration, is a leading figure in tumor transformation, including brain tumors. The acquisition of mutations in the mitochondrial DNA of tumor cells and the subsequent identification of the so-called mitochondria-related genes (MRGs), both functional (mutation of Complex I) and structural (mutations of Complex III/IV), have been seen to play an important role in metabolic reprogramming with increased proliferation, resistance to apoptosis, and the progression of tumorigenesis. This demonstrates that these mitochondrial alterations could have a role not only in the intrinsic tumor biology but also in the extrinsic one associated with the therapeutic response. We aim to summarize the main mitochondrial dysfunction interactions present in gliomas and how they might impact prognosis.

Phytochemical strategies in glioblastoma therapy: Mechanisms, efficacy, and future perspectives.

PMID: 39740382 | Indexed in PubMed

Glioblastoma (GBM) is foremost the most aggressive primary brain tumor, presenting extensive therapeutic challenges due to its high invasiveness, genetic complexity, and resistance to established treatments. Despite substantial advances in surgical and chemotherapeutic interventions, the median survival rate for patients is only 14.6 months, and the prognosis remains poor. This review focuses on the molecular hallmarks of GBM, including the activation of the PI3K/Akt pathway, genomic instability, and the deregulation of epidermal growth factor receptor (EGFR), all of which contribute to the tumor's aggressive behavior. Current therapies, such as Temozolomide and Bevacizumab, have limitations, highlighting the need for novel treatment strategies. Phytochemicals, bioactive compounds found in plants, have emerged as potential therapeutic agents by targeting multiple cellular pathways involved in GBM progression. This review provides an overview of key phytochemicals, including quercetin, curcumin, apigenin, and resveratrol. These compounds have shown promise in preclinical studies, with their anti-invasive, anti- angiogenic, pro-apoptotic, and anti-proliferative properties positioning them as strong candidates for GBM therapy. While phytochemicals offer a promising avenue for GBM treatment, further research is required to fully understand their mechanisms of action and to evaluate their efficiency in clinical settings. Developing multi-targeted, safer, and cost-effective anti-GBM therapies could significantly improve patient outcomes.

The impact of timing of temozolomide chemoradiotherapy for newly diagnosed

glioblastoma on patient overall survival: A multicenter retrospective study.

PMID: 39659837 | Indexed in PubMed

The optimal timing of initiating adjuvant temozolomide (TMZ) chemoradiotherapy after surgery in patients with glioblastoma is contentious. This study aimed to determine whether the timing of adjuvant treatment affects their overall survival (OS). Consecutive adult patients with histologically-confirmed newly diagnosed glioblastoma treated with adjuvant TMZ chemoradiotherapy across all neurosurgical centers in Hong Kong between 2006 and 2020 were analyzed. The surgery-to-chemoradiotherapy (S-CRT) interval was defined as the date of the first surgery to the date of initiation of adjuvant TMZ chemoradiotherapy. Four hundred and forty-one patients were reviewed. The median S-CRT interval was 40 days (interquartile range [IQR]: 33-47) and the median overall survival (mOS) was 16.7 months (95% CI: 15.9-18.2). The median age was 58 years (IQR: 50-63). Multivariable Cox regression with restricted cubic splines identified a nonlinear relationship between the S-CRT interval and mOS. Post hoc analysis-derived S-CRT intervals revealed that early CRT (9-12 weeks; aHR 1.07; 95% CI 0.67-1.71) were not significantly associated with OS. Subgroup analyses for the extent of resection (EOR) and pMGMT methylation status revealed no significant difference in treatment timing on OS. The timing of adjuvant TMZ chemoradiotherapy, if commenced within 12 weeks after glioblastoma diagnosis, did not influence OS regardless of EOR or pMGMT methylation status. Clinical judgment should be exercised in optimizing the timing of initiating adjuvant therapy.

Persistent Disparities in Survival for Patients with Glioblastoma.

PMID: 30149161 | Indexed in PubMed

Glioblastoma (GBM) is the most common malignant primary brain cancer in adults. Recent efforts have elucidated genetic features of tumor cells and thus enhanced our knowledge of GBM pathophysiology. The most recent clinical trials report median overall survival between 14 and 20 months. However, population level outcomes are quite variable and there is a paucity of such data within the literature. Three hundred seventy-two patients with GBM were diagnosed in the Canadian province of British Columbia between January 2013 and January 2015 and 278 patients had surgery. Of these, 268 had surgery in British Columbia and we have performed a retrospective review of their survival outcomes. Our results indicate a median age of 61.8 years at time of diagnosis, with a slight preponderance of male patients. The median overall survival was 10 months for patients in our cohort. As expected, patients older than the age of 65 and those with worse initial Karnofsky Performance Status scores had a poorer prognosis. Moreover, we have found extent of resection, treatment strategies, and treatment location affect overall survival. The present study highlights factors that affect patient survival after surgery in British Columbia. Our data are gathered within a single-payer, high-resource setting which removes possible confounders in outcome analysis. We find persistent differences in overall survival when compared with clinical trials and the Surveillance, Epidemiology, and End Results database. Further efforts should ensure access to the gold standard of care. All neuro-oncology centers should analyze the real-world outcomes of their local glioblastoma treatment strategies. Knowledge of the variance from expected and comparative results are fundamental for appropriate patient care.

Survival prognostic factors in patients with glioblastoma: our experience.

PMID: 22854592 | Indexed in PubMed

Approximate survival for glioblastoma is less than 1 year. Age, histological features and performance status at presentation represent the three statistically independent factors affecting longevity. The purpose of the study was to assess the role of surgery and to analyze prognostic factors in our patients operated for glioblastoma. We evaluated in 56 patients operated for glioblastoma their depressive and performance status in the preoperative and postoperative time. Moreover we analyzed the extent of surgery, the site and the size of lesions. Median overall survival was 17 months. An age of ≥60 years ($P<0.03$), a preoperative Karnofsky Performance Status KPS≤70 ($P=0.04$), a subtotal tumor resection (P5 cm ($P=0.01$), and no postoperative adjuvant treatment ($P=0.01$) were associated with the worst prognosis. Before surgery we found the presence of depression in 10 patients with a significative reduction of mean Back Depression Inventory scores after tumor resection ($P=0.03$). Finally, a KPS≤70 was significantly associated with an increased incidence of depression in the postoperative time. Tumor size, total resection and affective disorders were identified as predictors of survival in our series of patients with glioblastoma in addition to age and KPS score. In our opinion an early diagnosis and the use of specific safeguards in the operating room contribute to have an extension of the tumor progression time and median survival.

Preirradiation gemcitabine chemotherapy for newly diagnosed glioblastoma. A phase II study.

PMID: 11180090 | Indexed in PubMed

The median survival for patients with glioblastoma is reported to be 12 months. To improve the outcome for glioblastoma patients, the authors evaluated the therapeutic efficacy of preirradiation gemcitabine chemotherapy followed by standard radiotherapy. Twenty-one patients with newly diagnosed glioblastoma were enrolled in a prospective unicenter trial of preirradiation gemcitabine chemotherapy. Chemotherapy included up to 4 monthly cycles of intravenous gemcitabine (Day 1, Day 8, and Day 15; 1000 mg/m2). Involved field radiotherapy was given after chemotherapy or earlier in the case of disease progression or gemcitabine intolerance. With gemcitabine chemotherapy alone, there was a median progression free survival of 11 weeks and a progression free survival rate at 4 months of 24%. In 18 of 21 patients who subsequently received a full course of radiotherapy, the median progression free survival from the time of diagnosis was 8 months and the progression free survival rate at 12 months was 17% (3 of 18 patients). The median overall survival was 11 months. There was no specific treatment-related neurotoxicity reported. Neither age nor extent of residual postoperative tumor predicted the duration of progression free survival in patients treated with gemcitabine chemotherapy alone or in those treated with gemcitabine plus radiotherapy. Gemcitabine followed by radiotherapy is a safe regimen for patients with newly diagnosed glioblastoma but the gemcitabine schedule used in the current study did not appear to confer any survival advantage compared with standard involved field radiotherapy alone.